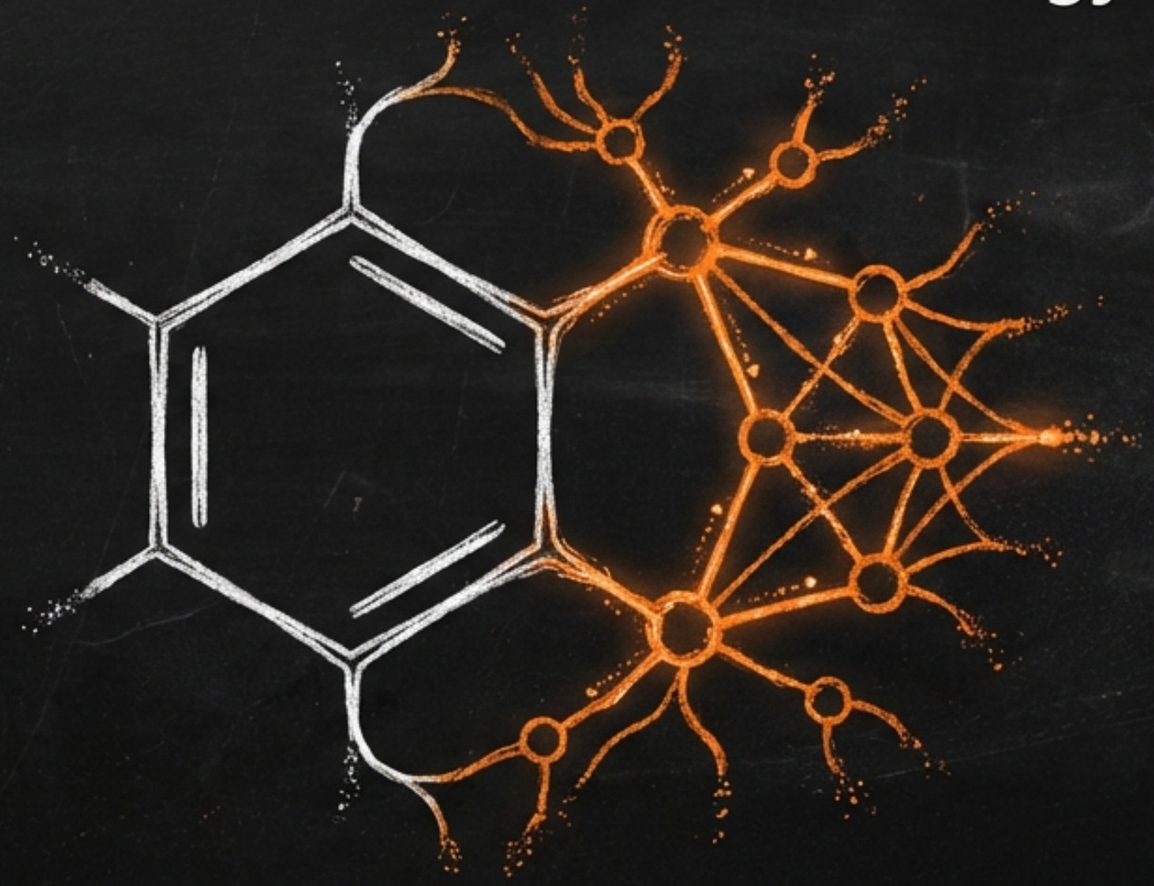
The New Colleague: Transforming Applied Chemistry with Generative AI

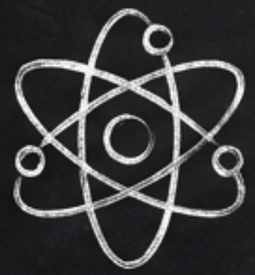
From Lab Bench to Data-Driven Strategy



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The Essential Bridge: Applied Chemistry as “Interface Lubrication Science”

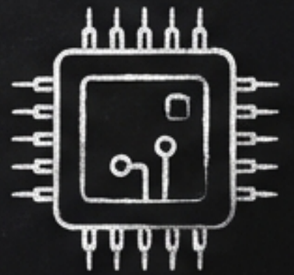
Applied chemistry provides the critical link between **fundamental scientific theory** and real-world industrial application. It addresses the technical challenges at the core of our key industries.



Basic Science



Interface Lubrication Science
(介面潤滑科學)



Industrial Application

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The Bedrock of Innovation: Core Chemical Training

Fundamental Disciplines

Ordinary Chemistry

Organic Chemistry

Physical Chemistry

Inorganic Chemistry

Analytical Chemistry

The Critical Role of Analysis

Rigorous experimental design and mastery of scientific instruments.

Direct Industry Link

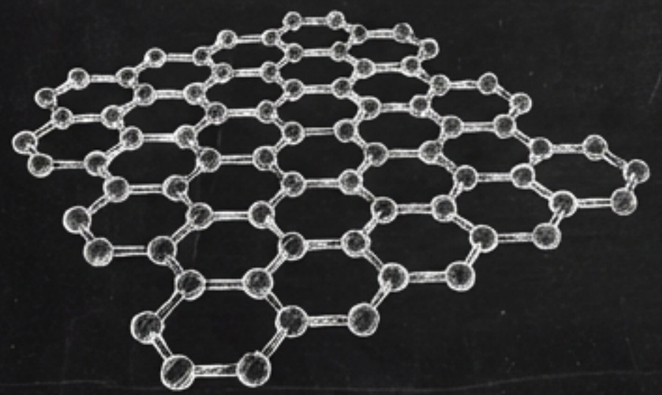
This training is directly correlated with high-tech manufacturing's demand for precision and **Yield Control (良率控制).**



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The Three Pillars of Modern Applied Chemistry Research

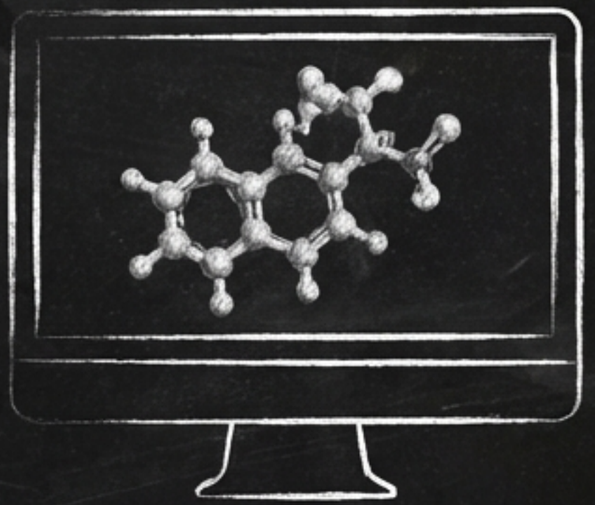
1. Materials Chemistry & Nanotechnology



2. Chemical Biology & Pharma Development



3. Computational Chemistry & Molecular Simulation

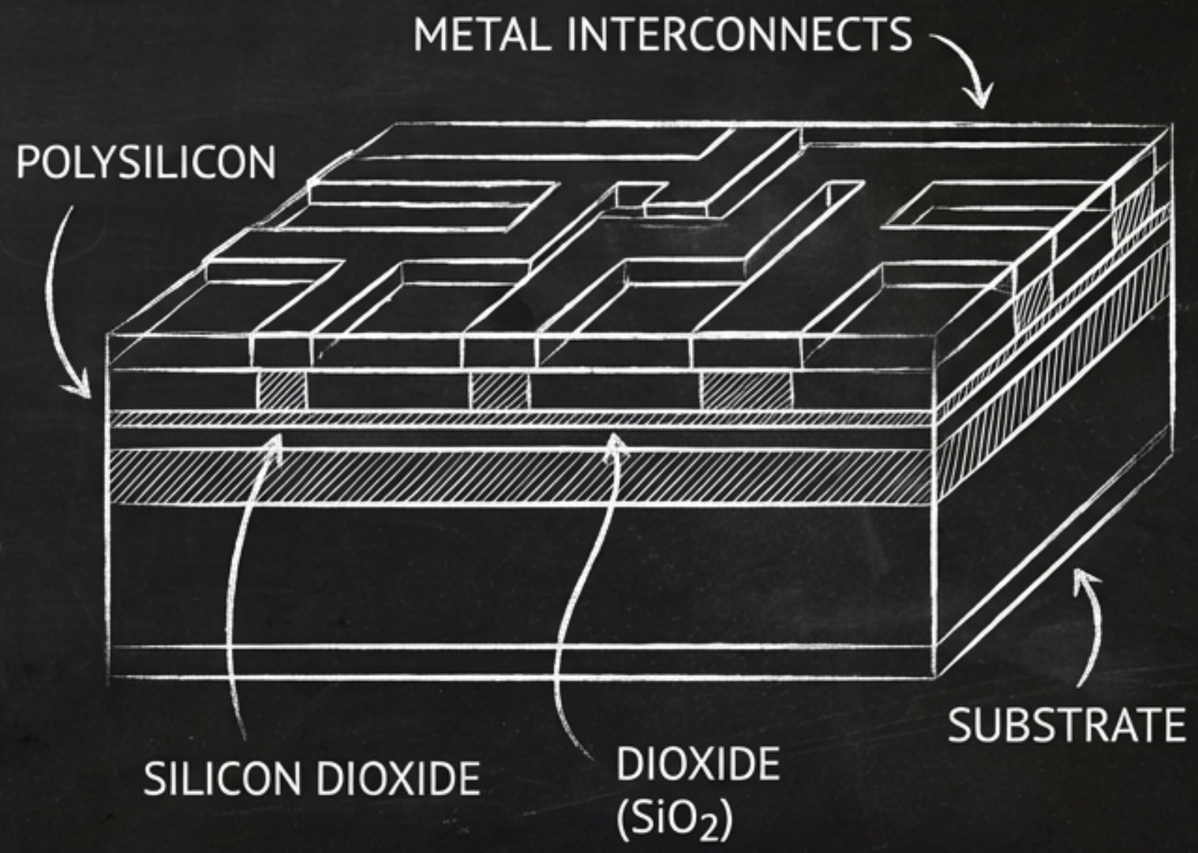


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1. Materials Chemistry & Nanotechnology

Engineering the Future of Electronics and Energy

- Developing next-generation materials for **photovoltaics, energy, and electronic components**.
- Mastering **inorganic nanomaterial synthesis** (metals, semiconductors).
- Meeting the demands of **semiconductor processes** and building supply chain resilience ('domestic substitution').

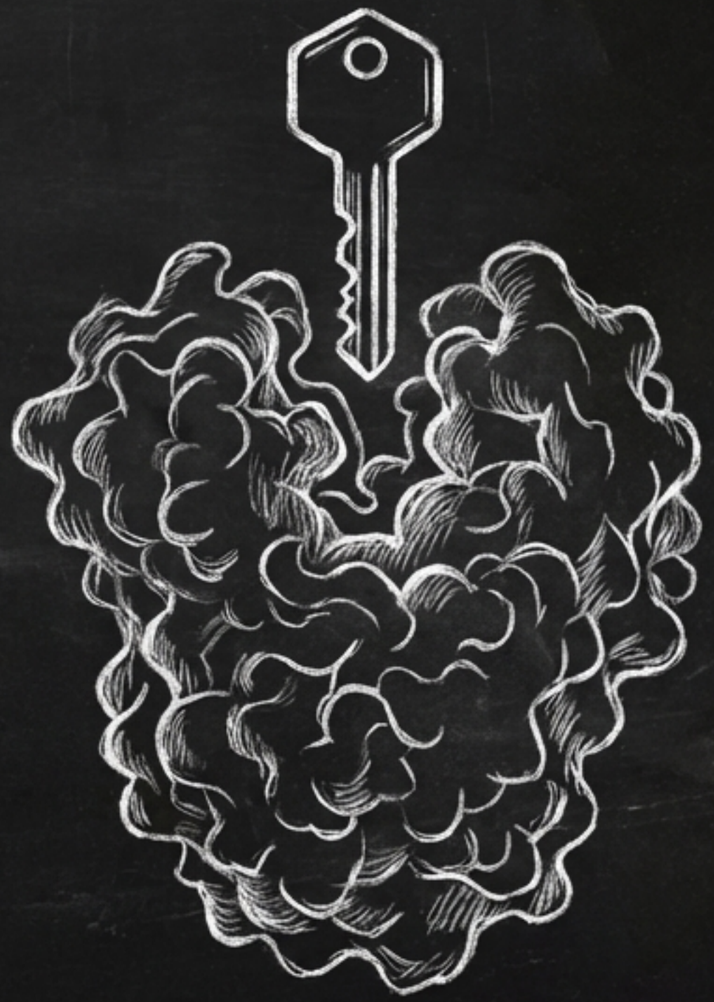


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2. Chemical Biology & Pharmaceutical Development

Solving Biological Puzzles with Chemical Tools

- Leveraging Medicinal Chemistry and Biocatalysis to address complex biological problems.
- The central challenge: The design and efficient synthesis of Active Pharmaceutical Ingredients (APIs).
- Optimizing synthesis processes for high-value industrial applications.



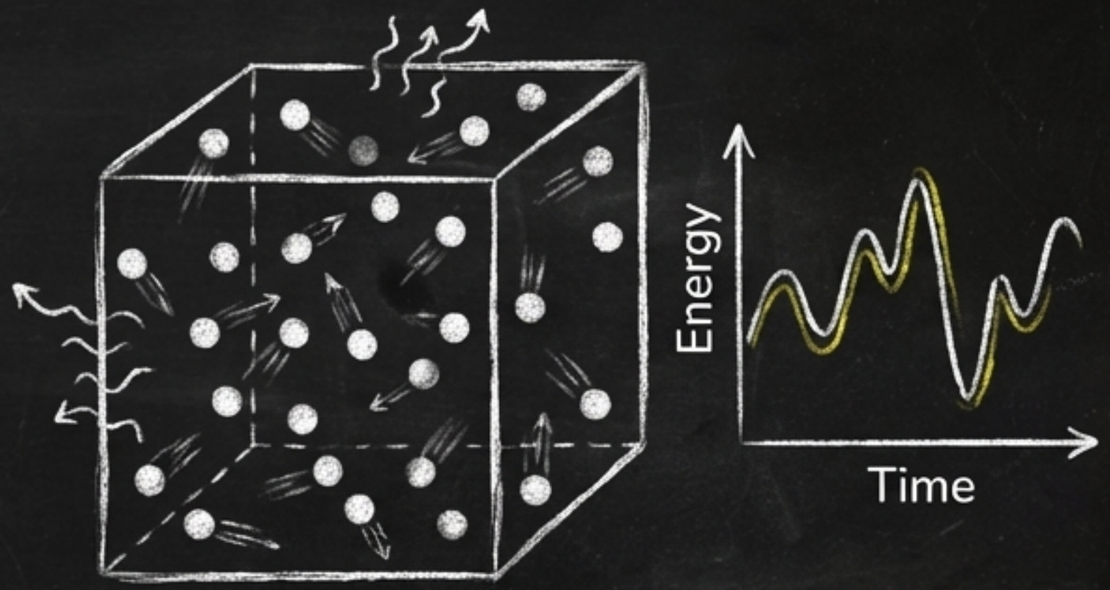
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3. Computational Chemistry & Molecular Simulation

Predicting Molecular Behavior Before a Single Experiment

- Key Methods:
  Ab initio calculations
  * Density Functional Theory (DFT)
  * Molecular Dynamics (MD)

Moving beyond time- and resource-intensive lab work by enabling virtual screening and property prediction. This is the foundation of ‘Rational Design.’

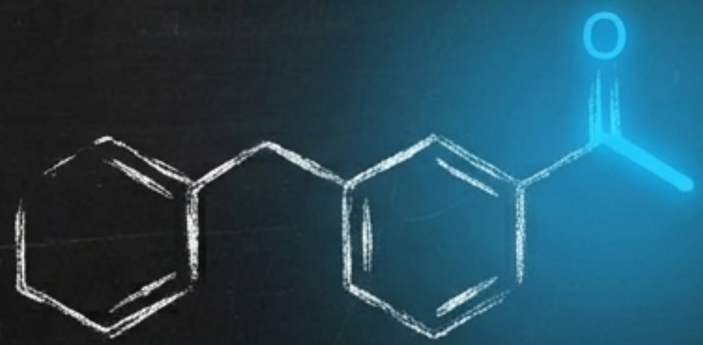


Rational Design (理性設計)

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Enter the Catalyst: Generative AI as a Collaborative Partner

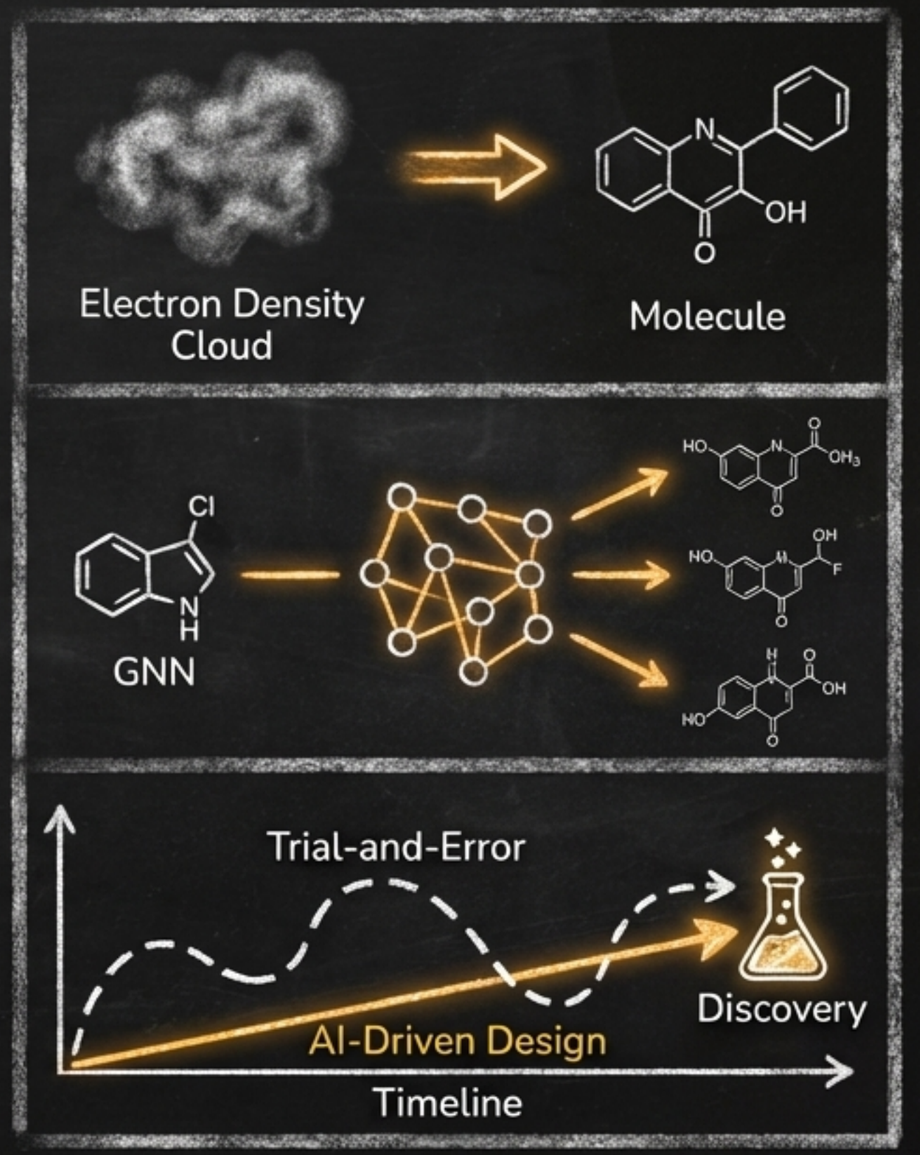
Generative AI is fundamentally transforming applied chemistry, elevating the role of the chemist from a traditional lab operator to a data-driven strategist (數據驅動的策略家).



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Transformation 1: Intelligent Molecular Design

- **Core Technology:** Intelligent Molecular Design Platforms using Graph Neural Networks (GNNs) and Genetic Algorithms (GAs).
- **Key Breakthrough:** The ED2Mol (Electron Density to Molecule) method. AI generates novel molecular structures based on quantum chemistry calculations of electron density.
- **Impact:** Massively accelerates innovation in organic light-emitting materials and drug design. Introduces 'Uncertainty Quantification' (UQ) to assess risk and focus resources.

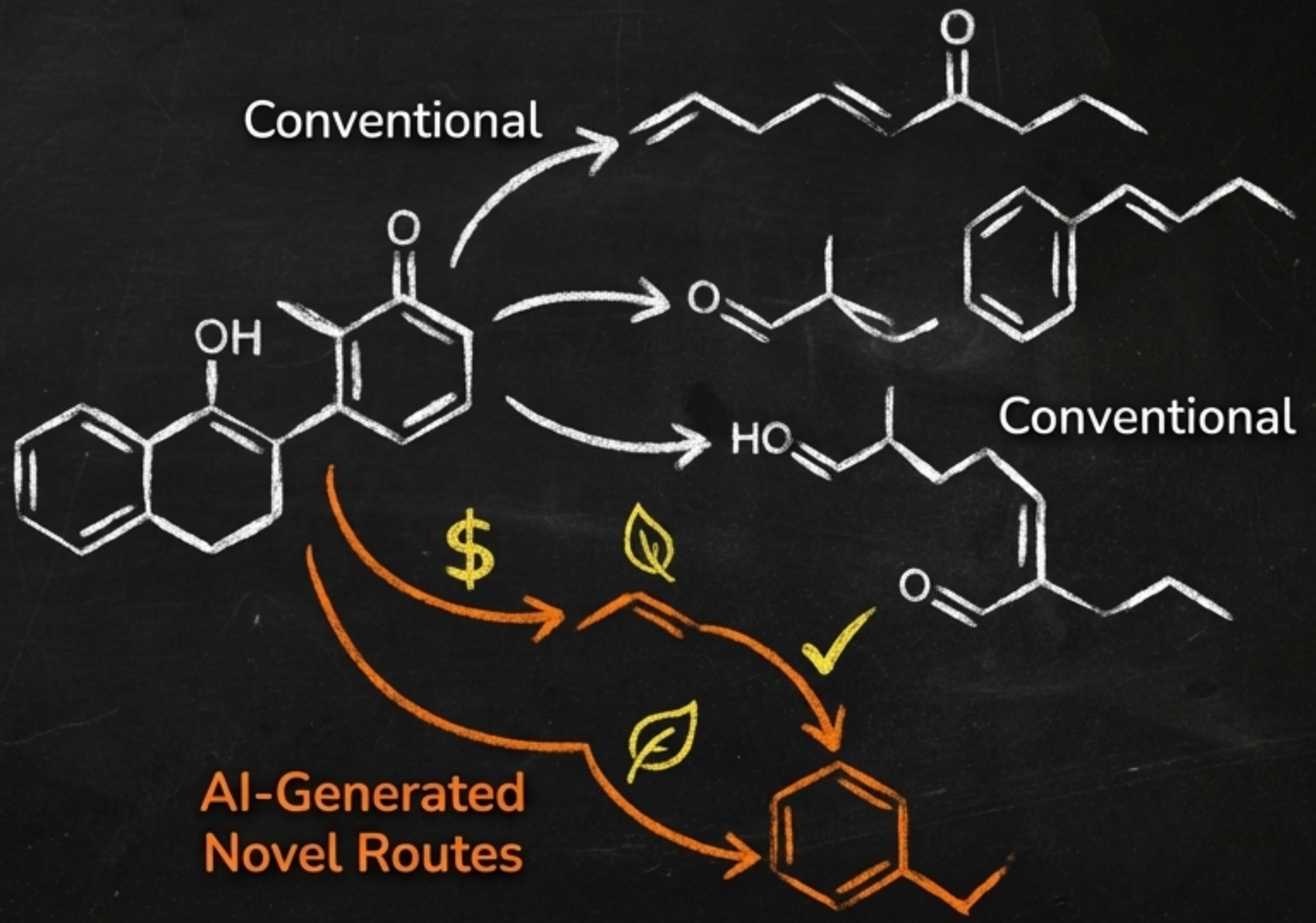


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Transformation 2: Automated Synthesis Planning (Retrosynthesis)

Unlocking **Novel, Faster,** and Sustainable Pathways

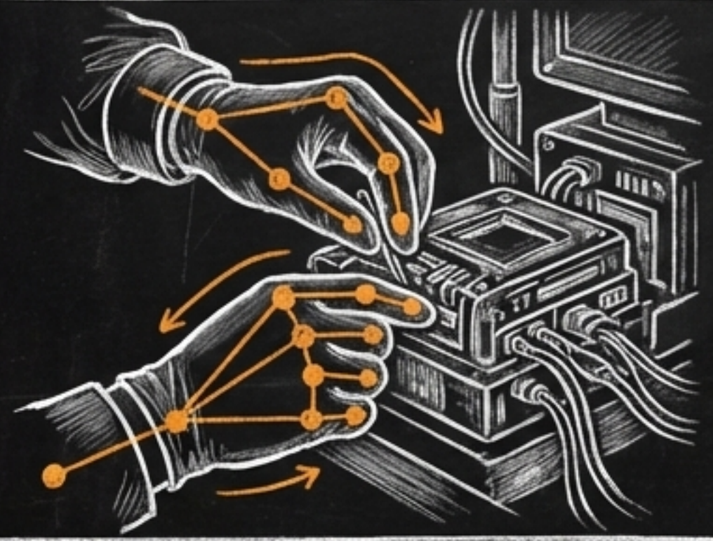
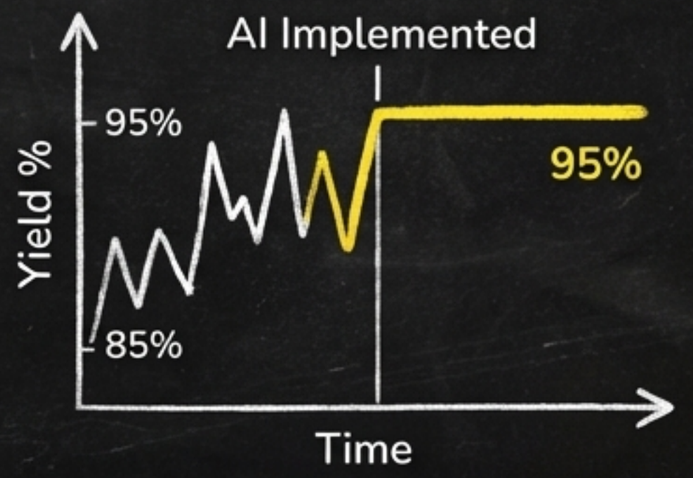
- Generates **diverse and innovative synthesis routes**, moving beyond conventional methods.
- Provides **feasibility assessments**, process chemistry optimization, and **impurity prediction**.
- Optimizes routes based on cost and sustainability: **commercial availability**, price, and path length.



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Transformation 3: Precision Manufacturing and Yield Enhancement

- **Application:** AI Vision in high-tech manufacturing (e.g., semiconductor fabrication).
- **Measurable Outcome:** Establishes a traceable process database to eliminate defects and **maintain and improve yield**.

Track Operations	Prevent Errors	Improve Yield
		

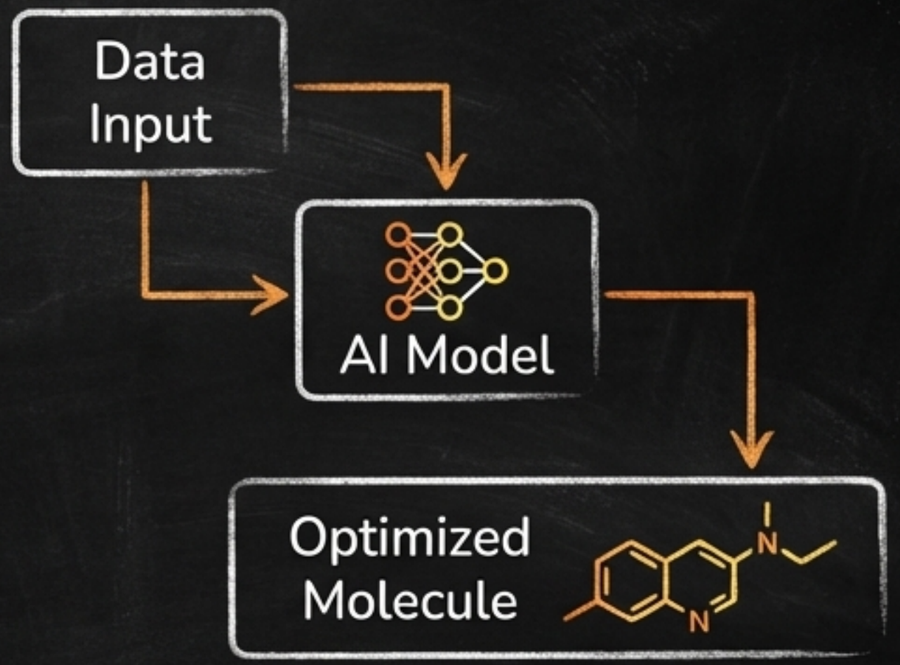
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The New Paradigm: From Trial-and-Error to Data-Driven Design

Trial-and-Error



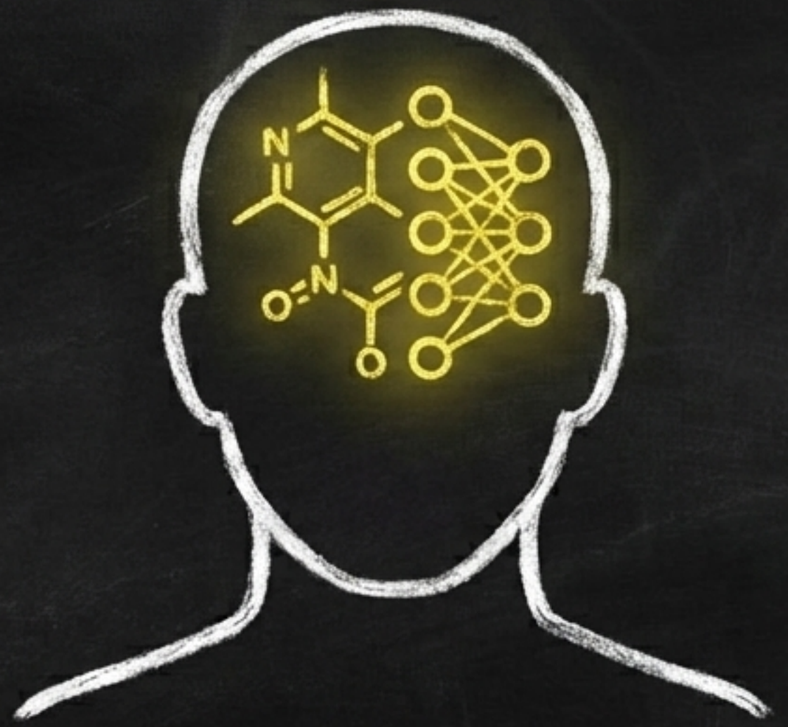
Data-Driven Rational Design



AI is the key collaborative partner for enhancing the efficiency and precision of applied chemistry, from laboratory discovery to factory-floor control.

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The Chemist, Evolved



No longer just a laboratory operator. The chemist of the future is a

<Data-Driven Strategist> (數據驅動的策略家),

who guides AI-powered discovery, interprets complex results,
and makes the critical decisions that drive innovation.

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